

# Multi-Agent Reinforcement Learning for Local Energy Market

Baptiste TARET supervised by Pedro FARIA<sup>1</sup> & Roman LE GOFF LATIMIER<sup>2</sup>

1. GECAD, School of Engineering (ISEP) at the Polytechnic Institute of Porto (IPP)
2. SATIE, ENS Rennes, Department of Mechatronics

**Keywords:** Multi-agent systems, Reinforcement learning, Local energy market, Distributed optimization, Price-based coordination, Congestion management

## Abstract

The transition to decentralized and renewable-based energy systems brings new challenges for the coordination of distributed actors in local electricity markets [1] (LEMs). This work investigates a multi-agent reinforcement learning (MARL) framework to enable intelligent, decentralized control of energy exchanges under physical network constraints. Each participant - producer, consumer, or prosumer - is modeled as an autonomous agent with local objectives and limited information, interacting via a peer-to-peer market. A distributed Alternating Direction Method of Multipliers (ADMM) algorithm allows agents to solve their optimization problems locally while achieving global consensus through message-based negotiation using the PEAK/SPADE platform and the XMPP communication protocol [2]. To manage grid congestion, we introduce a reinforcement learning agent trained using Soft Actor-Critic (SAC), which dynamically adjusts price signals to influence agents' decisions. Various reward shaping strategies are evaluated to improve constraint satisfaction, including continuous and anticipatory formulations. Results show that combining ADMM-based multi-agent optimization with RL-driven price modulation enables both realistic market behavior and effective congestion management [3]. This approach demonstrates the potential of MARL for aligning individual decisions with system-wide goals in future smart grid applications.

## 1 Introduction and Motivation

A simplified LEM is studied, it is composed of three agents - a producer (Agent 1), a consumer (Agent 2), and a Distribution System Operator (DSO, Agent 3) - each modeled as an autonomous entity with its own objectives and operational limits. A fourth agent, the Community Manager, oversees the synchronization of negotiations and enforces global consensus.

The study further explores how learning-based price signals can support congestion management. The Community Manager agent is trained using reinforcement learning to emit adaptive price signals, guiding individual agent decisions in a way that respects grid constraints.

## 2 Multi-Agent Optimization Framework

The optimization process is structured around alternating local and global phases, enabling agents to coordinate while operating asynchronously. After computing local exchange proposals based on their objectives and neighbor information, agents enter a synchronization protocol to ensure consistent progression.

Each agent sends its current state to its peers. Agents then wait to receive corresponding updates before proceeding. To avoid race conditions and incomplete information, a system of acknowledgements (ACKs) is implemented: agents confirm receipt of each other's messages before the next step can begin.

Once all agents have reached local convergence and exchanged ACKs, the Community Manager collects the full set of intended exchanges. It resolves inconsistencies through a consensus algorithm, and updates shared variables such as validated exchanges and prices. These are then broadcast to all agents, who resume their local computations based on the new signals.

This explicit synchronization protocol (Figure 1) ensures a globally coherent evolution of the distributed optimization, even under heterogeneous computation times and communication delays.

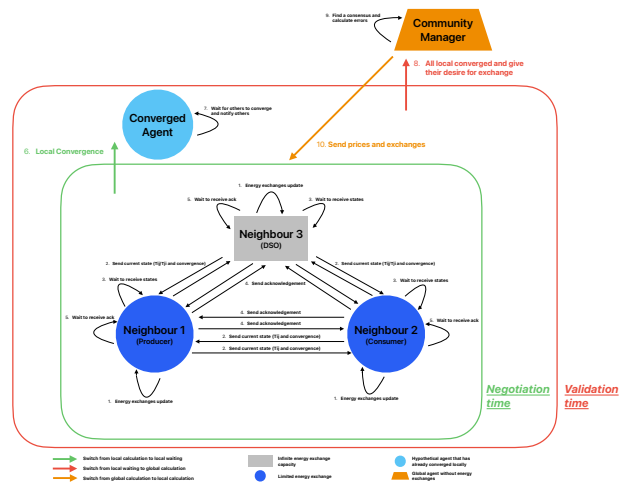


Figure 1. Multi-agent Energy Market Communication Process

### 3 Reinforcement Learning for Price-Based Congestion Management

To mitigate congestion at the interface with the DSO, a reinforcement learning agent is trained using the Soft Actor-Critic (SAC) algorithm. The Community Manager agent generates dynamic price signals, which guide local decisions throughout the convergence process and shape the resulting market exchanges.

#### 3.1 Reward Shaping Strategies

Two reward function formulations are evaluated:

- **Discontinuous Reward:** A two-part quadratic function: increasing and positive below the threshold  $P_l$ , decreasing and negative above.
- **Continuous Reward:** A smooth reward function shaped as a Gaussian is centered around the optimal value  $0.8P_l$ , providing gradual incentives and smooth gradients.



Figure 2. Comparison of Continuous and Discontinuous Reward Functions on Learning

Results presented in Figure 2 indicate that the Gaussian reward function provides the most stable learning behavior and achieves a 23.4% reduction in congestion compared to the baseline.

#### 3.2 Congestion Compliance Improvement

The introduction of reinforcement learning-based price signaling leads to a marked improvement in the system's ability to operate within network capacity limits.

Without price signaling, the DSO exchanged around 10 a.u. of power, far exceeding the network's limit. When the limit was set to 3 u.a and the agent was trained under this constraint, the exchanged power converged to approximately 3.33 a.u. While this still represents a slight violation of the limit (+11%), the overshoot is drastically reduced compared to the initial state (+233%). The introduction of reinforcement learning-based price signaling therefore yields an improvement of nearly 90% in respecting capacity constraints, significantly enhancing the system's ability to operate within network limits.

### 4 Integration into a Realistic Multi-Agent Setting

The RL agent is deployed in the ADMM-based multi-agent framework. When integrated into the ADMM-based multi-agent framework, the RL agent exhibits similar overall results in terms of limit compliance compared to the simple market case. However, the addition of a consensus phase with the Community Manager accelerates convergence towards the solution and modifies the trajectory by which it is reached (Figure 3). Because the RL agent's observations now include the intermediate consensus state, its perception of the environment changes, leading to different actions being selected. As a result, the final operating point in the multi-agent setting does not converge to the exact same value as in the simple market scenario, even though the performance in respecting capacity limits remains comparable.

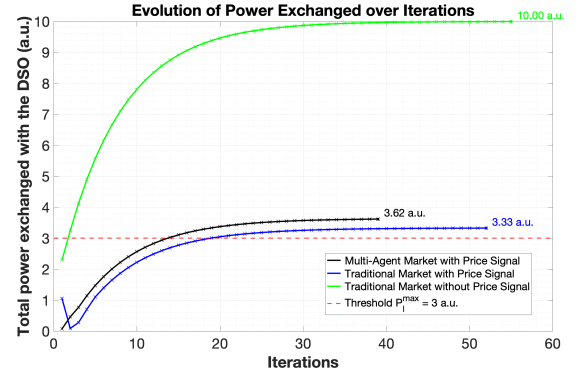


Figure 3. Comparison of Continuous and Discontinuous Reward Functions on Learning

The same RL agent was then applied to multiple markets where agents' demands slightly differed from the training conditions, in order to place it in more realistic conditions. While performance was generally lower, the learning proved robust. Training the same agent on varying markets did not improve its adaptability to changing conditions, a limitation attributed to the inherent structure of the ADMM-based resolution.

### 5 Conclusion and Outlook

The integration of ADMM-based multi-agent coordination with reinforcement learning has been shown to enable effective and realistic congestion management in local energy markets. As a next step, a more realistic setting will be considered where price signals are only applied to exchanges with the DSO to manage grid congestions. Future extensions of the framework will include a battery agent, trained to dynamically manage charging and discharging, in order to reduce DSO-related costs. The coordinated operation of multiple RL agents will be investigated to further enhance system flexibility and performance.

## Acknowledgements

Many thanks to Roman LE GOFF LATIMIER, Pedro FARIA and GECAD's team for warmly welcoming me. They guided me during this internship and honoring me with their precious knowledge.

## References

- [1] Hien Thanh Doan et al. "Fully Decentralized Peer-to-Peer Community Grid With Dynamic and Congestion Pricing". In: *IEEE Internet of Things Journal* 11.14 (2024), pp. 24483–24496. DOI: 10.1109/JIOT.2024.3379746.
- [2] Bruno Ribeiro et al. "A Novel Framework for Multiagent Knowledge-Based Federated Learning Systems". In: *Advances in Practical Applications of Agents, Multi-Agent Systems, and Cognitive Mimetics. The PAAMS Collection*. Ed. by Philippe Mathieu et al. Cham: Springer Nature Switzerland, 2023, pp. 296–306. ISBN: 978-3-031-37616-0.
- [3] Guénolé Chérot et al. "Learning-Based Optimization of the Exchange Price in a Heterogeneous Market". In: *2024 22nd International Conference on Intelligent Systems Applications to Power Systems (ISAP)*. 2024, pp. 1–7. DOI: 10.1109/ISAP63260.2024.10744279.